



Graph Algorithms and Applications

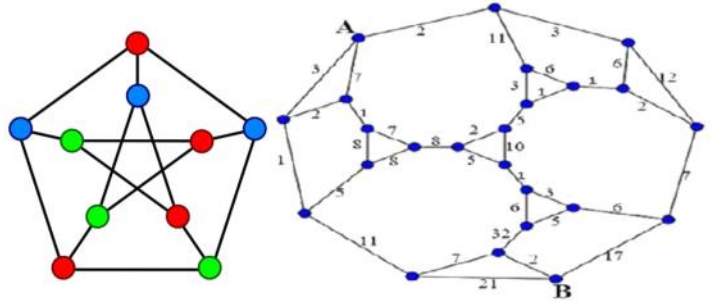
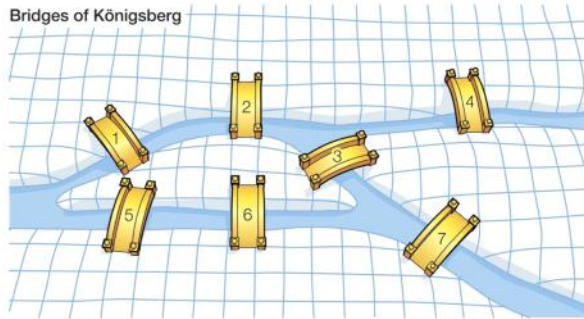
Course Code: 18B14CI645

Credit: 3

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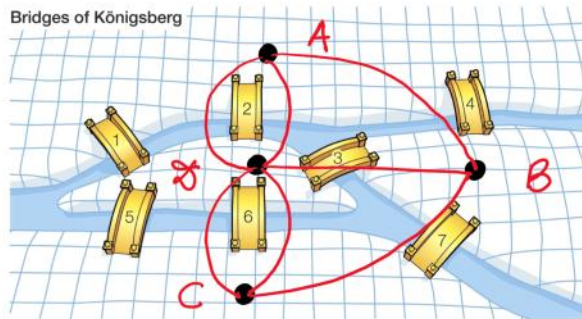
Bridges of Königsberg



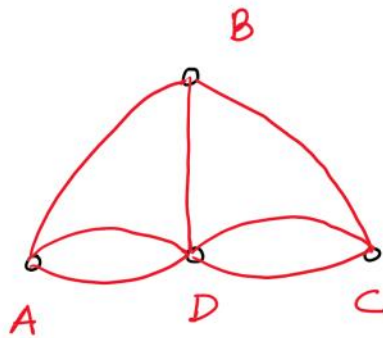
Graph Theory: The Language of Connections

Euler Trail and Circuit

Vertices & Edges are the most fundamental in graph theory.



what is this?



Q:- Is it possible to start at a vertex, go to every edge exactly once and come back to the same vertex?

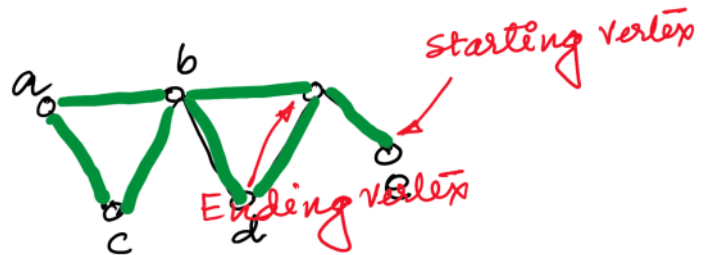
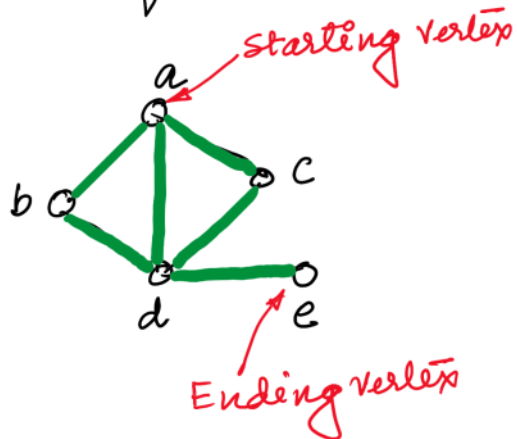
A:- NO why?

Because this does not have Euler's circuit.

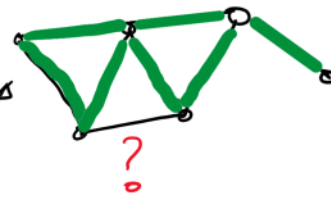
Hence, the title of the lecture is Euler's trail & circuit.

An Euler trail is a trail that visits every edge exactly once. "Sometimes also called as Euler's path"

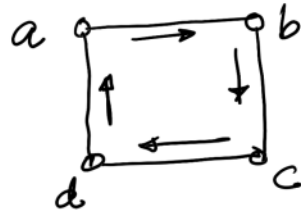
Check for the Euler's Trail in the foll. graphs:-



This graph does not have Euler trail.



A closed Euler trail is called an Euler circuit.



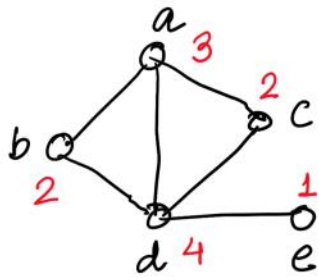
How can you check a given graph has Euler trail or not? (without drawing it in a graph)

It is related to the deg. of vertices.

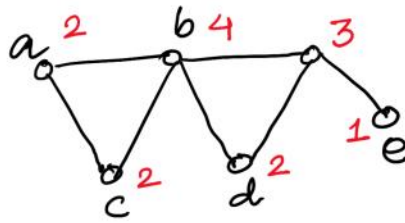
If all the vertices are of even degrees then the graph has Euler circuit.

If two vertices are of odd degree then the graph has Euler trail but not circuit

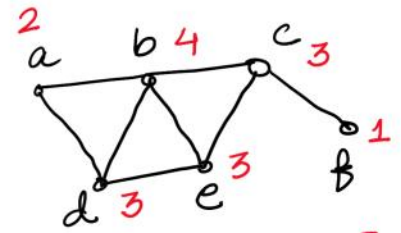
If more than two vertices are of odd degree then the graph has neither Euler trail nor Euler circuit.



$[1, 2, 2, 3, 4]$
 ↑ ↑
 2 odd deg.
 Euler trail

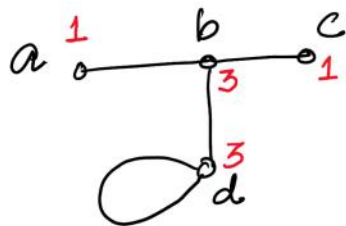


$[1, 2, 2, 2, 3, 4]$
 ↑ ↑
 2 odd deg.
 Euler trail



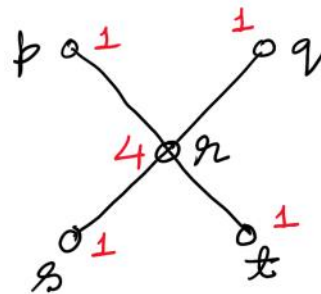
$[1, 2, 3, 3, 3, 4]$
 ↑ ↑ ↑ ↑
 4 odd deg.
 No Euler trail

use deg. method to check whether Euler trail exist or not.



$[1, 1, 3, 3]$

↑
More than 2 odd
No Euler trail



$[1, 1, 1, 1, 4]$

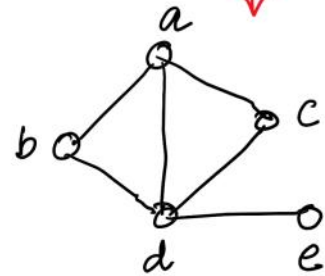
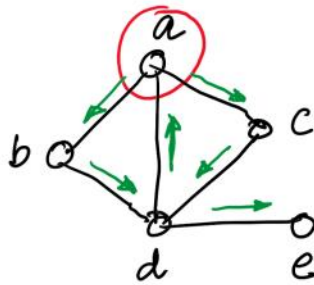
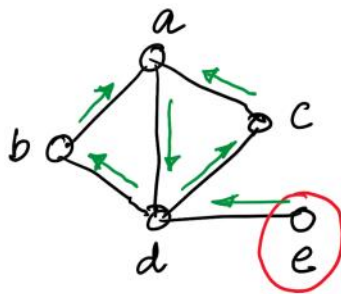
↑
More than 2 odd
No Euler trail

If it get confirmed that the grap has an Euler trail with 2 odd degrees then

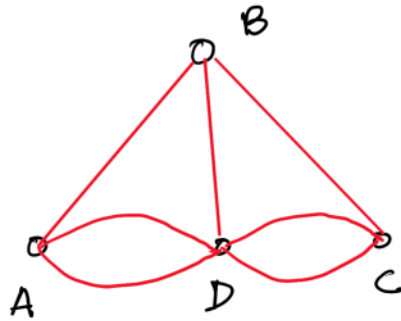


Start with odd deg. always.

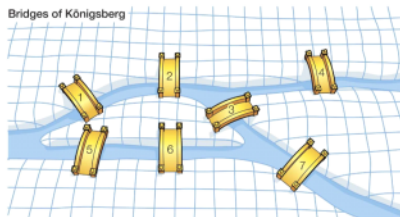
Try with even deg.



check for Euler's circuit

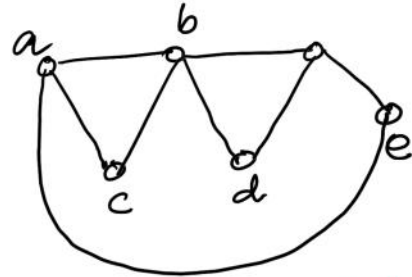
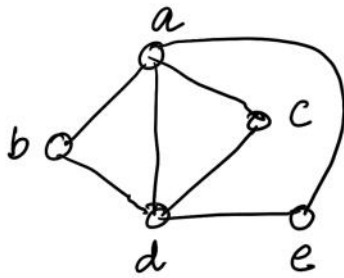


Does not have Euler circuit.



← Hence, does not have any solⁿ.

check for Euler circuit in the foll. graphs:



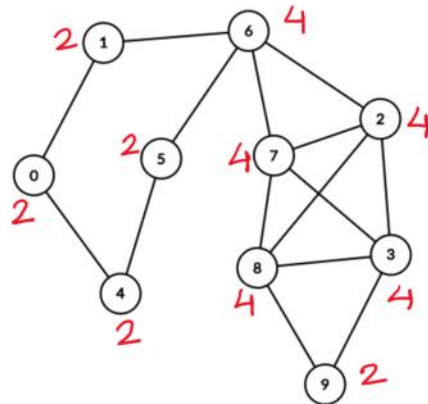
For this the graph has to be connected.

If a graph has a Euler circuit in it then that graph is called as Eulerian.

Or

A graph is called Eulerian if and only if all vertices in it are even.

Is the following graph Eulerian?



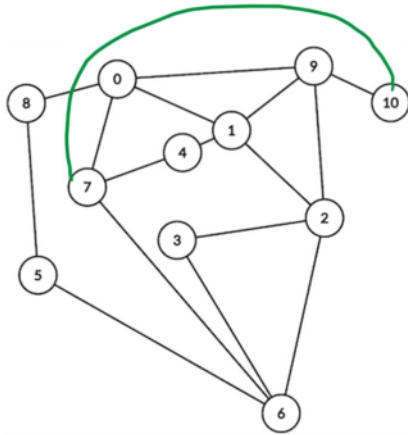
All the deg.'s are
Even
Hence, Eulerian.

In General, a graph has a Euler trail if the degree of at least two vertices is odd.

TRUE

FALSE ✓

For the graph below, adding an edge between vertex 7 and 10 will make the graph Eulerian.



why?
 $\deg(7) = 3$
 $\deg(10) = 1$